# Billable work review

23 October - 9 November 2026 | South Africa time (SAST)

For Paul to review before invoicing. No invoice has been created, changed or sent.

| Project | Reviewed work | Tensology calendar | Difference |
|---|---|---|---|
| Jackcat | 01:41:50.651116 | 02:57:50.651116 | 01:16:00 |
| Merrypak | 03:38:26.798153 | 04:48:29.105035 | 01:10:02.306882 |
| Multisnack (R0) | 00:47:46.373861 | 00:47:46.373861 | 00:00:00 |
| Player1Sport | 01:59:30.383 | 01:59:30.383 | 00:00:00 |
| Snuza | 04:32:50.083269 | 04:36:03.819227 | 00:03:13.735958 |
| Spoortrack [new] | 02:41:16.051908 | 04:07:16.051908 | 01:26:00 |

Reviewed billable work: 14:33:53.967446. Multisnack tracked separately at R0: 00:47:46.373861.
The preserved calendar entries add 03:55:16.04284 beyond the IDE-reviewed time. Review the differences section before using the calendar total for an invoice.
All 94 billable/R0 sync entries are active in Tensology. Seven internal or routine entries are retained as non-billable. The 8 existing entries were preserved exactly as requested. Overlapping local source records are counted once.
Durations and start/end times retain sub-second precision. No quarter-hour or half-hour rounding has been applied. Project hours can overlap between projects; this document shows the sum of the reviewed time for each project.
Multisnack belongs on the Crystal Logic invoice as struck-through work at R0. Routine automated runs and projects you designated non-billable are excluded.
Review order: project details follow alphabetically, then the eight preserved calendar entries and their overlap differences.

## Jackcat

| Date | Start / end (SAST) | Work performed | Duration |
|---|---|---|---|
| 23 Oct 2026 | 10:32:17.229585<br>10:33:54.399245 | Investigated the catalogue 500 errors and traced the cause to a search context processor that fails when the Redis cache is unreachable, and outlined options to restore or harden the connection. | 00:01:37.16966 |
| 23 Oct 2026 | 10:34:08.575101<br>10:34:22.639786 | Confirmed the plan for the Redis catalogue outage: check the cache service, correct its connection settings, verify the catalogue page loads, and ensure error pages still render when the cache is down. | 00:00:14.064685 |
| 23 Oct 2026 | 10:34:36.017183<br>10:38:48.585424 | Investigated catalogue errors during cache outages and added error handling around the search-page cache. | 00:04:12.568241 |
| 23 Oct 2026 | 10:39:00.595456<br>10:41:07.814842 | Review required: independent review of the Redis outage fix found other catalogue data lookups still unprotected, so the change was judged incomplete and not ready to ship. | 00:02:07.219386 |
| 23 Oct 2026 | 10:44:23.56317<br>10:47:38.93577 | Review required: second review pass on the Redis outage fix again reported unprotected cache lookups in catalogue listing and part detail code, with no verification possible while Redis was down. | 00:03:15.3726 |
| 23 Oct 2026 | 10:47:58.309221<br>10:54:41.562353 | Investigated Redis failure handling in catalogue functions and attempted database fallback changes; validation was blocked by the local Django environment. | 00:06:43.253132 |
| 23 Oct 2026 | 11:03:51.671327<br>11:07:54.501394 | Extended cache error handling in catalogue manufacturer, publishing-group and exclusion queries, and checked the updated Python syntax. | 00:04:02.830067 |
| 23 Oct 2026 | 16:58:15.358601<br>17:02:52.008707 | Mapped the Redis-dependent catalogue page and error page paths to plan the fix for the connection failure causing 500 errors, and set out steps to restore Redis and make error pages render without it. | 00:04:36.650106 |
| 23 Oct 2026 | 17:03:09.768756<br>17:03:33.107805 | Prepared a written execution contract for the Jackcat catalogue outage, defining the Redis recovery and error-page fallback scope, what was excluded, and the end-to-end user journey to verify. | 00:00:23.339049 |
| 23 Oct 2026 | 17:03:46.492186<br>17:07:57.578345 | Configured cache connection timeouts and strengthened catalogue error-page handling for cache outages. | 00:04:11.086159 |
| 23 Oct 2026 | 17:08:15.4269<br>17:12:36.96847 | Reviewed catalogue cache settings and error-page handling to assess outage resilience; the review stopped before a final verdict. | 00:04:21.54157 |
| 23 Oct 2026 | 17:12:50.440523<br>17:18:58.706223 | Re-validated the catalogue Redis outage fix: confirmed cache settings and database fallbacks prevent 500 errors when Redis is unreachable, and proved cache reads return misses instead of raising. | 00:06:08.2657 |

## Jackcat

| Date | Start / end (SAST) | Work performed | Duration |
|---|---|---|---|
| 23 Oct 2026 | 17:19:16.741946<br>17:23:24.980282 | Independently reviewed the catalogue Redis outage fix, checking that cache failures fall back to the database and that the error page no longer re-triggers a Redis failure. | 00:04:08.238336 |
| 23 Oct 2026 | 17:23:42.011142<br>17:30:46.390715 | Re-ran validation of the Jackcat catalogue Redis outage fix: confirmed Redis is healthy, proved the cache-outage path no longer raises errors, and inspected the settings, error-handler and catalogue code, but full tests and browser checks were blocked by an incomplete local environment. | 00:07:04.379573 |
| 23 Oct 2026 | 17:31:01.750513<br>17:35:38.782479 | Re-ran checks for the Redis/catalogue 500 fix: confirmed Redis responds, Django system check is clean, and cache-resilience tests pass, showing the catalogue page loads with Redis up or down. | 00:04:37.031966 |
| 23 Oct 2026 | 17:35:55.299858<br>17:38:22.060901 | Independently reviewed the Redis/catalogue 500 fix, checking that cache failures no longer break the catalogue or error pages, running the test suite and system checks, and flagging pre-existing security concerns for follow-up. | 00:02:26.761043 |
| 23 Oct 2026 | 17:38:36.375597<br>17:40:44.683519 | Re-ran all validation for the Redis/catalogue 500 fix, confirming a clean system check, all 19 tests passing, and that the catalogue and error pages keep working when Redis is unavailable. | 00:02:08.307922 |
| 23 Oct 2026 | 17:40:59.661074<br>17:44:02.644523 | Independent review of the fix for Redis connection failures causing catalogue page errors, checking the caching configuration, error-page fallbacks, and new resilience tests. | 00:03:02.983449 |
| 23 Oct 2026 | 17:44:17.304596<br>17:47:01.238113 | Re-ran the catalogue and cache-resilience tests plus system checks and Redis connectivity to confirm the fix holds; no defects found and no code changes made. | 00:02:43.933517 |
| 23 Oct 2026 | 17:47:16.226364<br>17:49:10.188214 | Independent review of the Redis failure fix, verifying the cache configuration, database fallbacks across catalogue and middleware code, and error-page safety net, with follow-up advisories noted. | 00:01:53.96185 |
| 23 Oct 2026 | 17:49:24.668606<br>17:50:56.642652 | Reviewed the Redis cache resilience fix for the catalogue pages, re-ran the six cache resilience tests and the Django system check, and confirmed the cache and error-handling changes were intact with no defects to correct. | 00:01:31.974046 |
| 23 Oct 2026 | 17:51:11.307805<br>17:51:47.729776 | Prepared the final ticket report summarising the Redis cache resilience fix, its verification results, remaining risks and recommended follow-up work, and recorded durable project notes for future runs. | 00:00:36.421971 |
| 23 Oct 2026 | 18:25:14.637877<br>18:26:58.861149 | Investigated the catalogue 500 errors caused by Redis being unreachable, mapping the affected catalogue routes, cache lookups and error-handling paths, and outlined a plan to restore Redis and harden the fallback. | 00:01:44.223272 |

## Jackcat

| Date | Start / end (SAST) | Work performed | Duration |
|---|---|---|---|
| 23 Oct 2026 | 18:27:13.207092<br>18:27:36.215959 | Prepared a written execution plan for the Jackcat ticket investigating Redis connection failures that cause 500 errors on catalogue pages, covering diagnosis steps, test coverage and scope boundaries. | 00:00:23.008867 |
| 23 Oct 2026 | 18:27:50.030847<br>18:35:31.979855 | Investigated catalogue page errors caused by Redis being unreachable, confirmed Redis was running and no local settings overrode the cache, then added cache failure tolerance and timeouts plus tests covering the catalogue page and search form fallback. | 00:07:41.949008 |
| 23 Oct 2026 | 18:35:51.582226<br>18:35:52.138 | Independently reviewed Jackcat's Redis cache-failure fix by inspecting the diff, settings overrides and new resilience tests, and ran the catalogue regression tests, which passed. | 00:00:00.555774 |
| 23 Oct 2026 | 18:35:52.138<br>18:37:42.236 | Independently reviewed the Jackcat catalogue Redis connection fix by inspecting the changed cache and settings code and running the catalogue resilience test suite. | 00:01:50.098 |
| 23 Oct 2026 | 18:37:42.236<br>18:38:24.881052 | Independently reviewed Jackcat's Redis cache-failure fix by inspecting the diff, settings overrides and new resilience tests, and ran the catalogue regression tests, which passed. | 00:00:42.645052 |
| 02 Nov 2026 | 10:08:08.348372<br>10:13:54.482724 | Fixed catalogue pages returning server errors when a non-numeric vehicle identifier was used, so invalid vehicle and part addresses now show a proper not-found page, with regression tests added and verified. | 00:05:46.134352 |
| 02 Nov 2026 | 10:19:05.050982<br>10:22:31.734265 | Fixed the catalogue part page so an unknown part ID returns a not-found response instead of a server error, and added regression tests covering unknown, encoded and valid part URLs. | 00:03:26.683283 |
| 02 Nov 2026 | 10:22:41.638298<br>10:24:32.224855 | Independently reviewed the catalogue unknown-part-ID change against its acceptance criteria, confirming the not-found behaviour and test suite pass, and noting that two new tests also pass on the pre-change code. | 00:01:50.586557 |
| 02 Nov 2026 | 10:25:54.209843<br>10:25:55.814 | Reviewed the Jackcat catalogue part page's handling of unknown part IDs, inspecting the existing unverified 404 fix and regression test, and stopped for approval before implementing or testing. | 00:00:01.604157 |
| 02 Nov 2026 | 10:25:55.814<br>10:26:25.243 | Reviewed the Jackcat catalogue ticket about unknown part IDs causing server errors, inspecting the existing candidate fix and regression test without making changes. | 00:00:29.429 |
| 02 Nov 2026 | 10:26:25.243<br>10:26:30.67779 | Reviewed the Jackcat catalogue part page's handling of unknown part IDs, inspecting the existing unverified 404 fix and regression test, and stopped for approval before implementing or testing. | 00:00:05.43479 |

## Jackcat

| Date | Start / end (SAST) | Work performed | Duration |
|---|---|---|---|
| 02 Nov 2026 | 10:26:36.339247<br>10:26:37.341 | Verified that unknown catalogue part IDs now return a not-found response while valid part pages still display, and strengthened the regression tests covering missing part records. | 00:00:01.001753 |
| 02 Nov 2026 | 10:26:37.341<br>10:29:02.977 | Verified that unknown part IDs return a 404 page instead of a server error, and strengthened the regression tests covering missing and slash-encoded part IDs. | 00:02:25.636 |
| 02 Nov 2026 | 10:29:02.977<br>10:29:06.763214 | Verified that unknown catalogue part IDs now return a not-found response while valid part pages still display, and strengthened the regression tests covering missing part records. | 00:00:03.786214 |
| 02 Nov 2026 | 11:42:25.480411<br>11:42:28.024 | Verified the Jackcat catalogue fix so unknown part IDs return a not-found page instead of a server error, while valid part pages still display correctly. | 00:00:02.543589 |
| 02 Nov 2026 | 11:42:28.024<br>11:44:23.315 | Audited and tested the Jackcat catalogue change covering unknown part IDs, valid part pages and malformed vehicle IDs; no further code changes were needed. | 00:01:55.291 |
| 02 Nov 2026 | 11:44:23.315<br>11:44:26.107443 | Verified the Jackcat catalogue fix so unknown part IDs return a not-found page instead of a server error, while valid part pages still display correctly. | 00:00:02.792443 |
| 02 Nov 2026 | 14:22:29.328153<br>14:23:39.22213 | Independently reviewed the catalogue fix for unknown part IDs: confirmed unknown parts return a not-found page and valid parts still render, and flagged that the URL-level regression test is untracked. | 00:01:09.893977 |
| **TOTAL** | | | **01:41:50.651116** |

## Merrypak

| Date | Start / end (SAST) | Work performed | Duration |
|---|---|---|---|
| 24 Oct 2026 | 21:15:34.36<br>21:40:40.728 | Investigated the Merrypak production host-header errors, tracing the localhost requests to a direct call to the application server rather than through the web proxy, and checked server logs and deploy scripts for the source. | 00:25:06.368 |
| 25 Oct 2026 | 12:43:48.203<br>12:48:42.96 | Added a validation tool for order-file formats and documented their D3 integration mapping. | 00:04:54.757 |
| 26 Oct 2026 | 15:36:37.652975<br>15:38:53.556705 | Investigated the Merrypak admin settings page error, tracing it to a removed Django template filter and confirming the project's own templates already use a compatible replacement. | 00:02:15.90373 |
| 26 Oct 2026 | 15:39:42.709578<br>15:41:05.807625 | Reviewed the admin settings page failure and narrowed the remaining risk to a third-party admin package's templates plus a Django version mismatch between the ticket and the project. | 00:01:23.098047 |
| 26 Oct 2026 | 15:41:23.415798<br>15:42:11.395096 | Defined the fix plan for the admin settings page error, listing the templates to check, the test to keep, and the condition for confirming the page loads. | 00:00:47.979298 |
| 26 Oct 2026 | 15:42:27.284141<br>15:46:28.61878 | Extended the admin template compatibility test to cover the settings page and added a check that simulates the newer Django version; the test could not be run in this session. | 00:04:01.334639 |
| 27 Oct 2026 | 09:47:15.445266<br>09:50:24.935841 | Investigated the Merrypak admin settings page error caused by an outdated template filter, confirming the leftover came from a third-party admin package already overridden by project templates. | 00:03:09.490575 |
| 27 Oct 2026 | 09:50:45.441681<br>09:54:29.00728 | Reviewed the Merrypak admin template fix and its regression test, checking that project overrides replace the outdated filter and that package versions were left unchanged; the test run was blocked. | 00:03:43.565599 |
| 27 Oct 2026 | 09:54:44.75695<br>09:56:12.246066 | Corrected the Merrypak admin template compatibility test by adding the missing parent template to the scanned chain; the test run itself could not be executed in this session. | 00:01:27.489116 |
| 27 Oct 2026 | 10:11:12.731651<br>10:13:36.226964 | Re-checked the Merrypak admin settings page templates for the removed Django filter that caused the 500 error, confirming the project's own templates no longer use it and that the test file covers the admin change page. | 00:02:23.495313 |
| 27 Oct 2026 | 16:44:49.797846<br>16:46:02.809511 | Investigated the Merrypak Django admin settings page failure caused by the removed length_is template filter, inspecting the diff, test file and templates, but shell access was blocked so the scoped test could not be rerun and no fix was applied. | 00:01:13.011665 |
| 28 Oct 2026 | 10:52:35.639054<br>10:54:57.162929 | Investigated the recurring Celery worker-dead alert on jackcat-ub-srv; server access was blocked, so only repository configuration was reviewed and a possible broker vhost mismatch was noted as unproven. | 00:02:21.523875 |

## Merrypak

| Date | Start / end (SAST) | Work performed | Duration |
|---|---|---|---|
| 28 Oct 2026 | 10:59:16.408896<br>11:00:32.785992 | Continued investigating the recurring Celery worker-dead alert for the Merrypak site, comparing the production worker's broker settings with the startup script, but live server checks could not run because shell access was blocked. | 00:01:16.377096 |
| 28 Oct 2026 | 11:00:43.259361<br>11:02:03.051508 | Continued the Merrypak Celery worker-dead investigation by reviewing the project's Celery and RabbitMQ configuration files, forming a hypothesis that the worker and monitor use different broker vhosts; live server checks were blocked. | 00:01:19.792147 |
| 28 Oct 2026 | 11:53:05.426375<br>11:54:01.575408 | Investigated the recurring Celery worker-dead alert on the Merrypak server by reviewing the service and broker configuration files, but live SSH checks were blocked, so the worker ping and broker permissions remain unverified. | 00:00:56.149033 |
| 28 Oct 2026 | 11:54:21.277152<br>11:56:06.631059 | Independent review of the Celery worker-dead work found no ticket-related code change and unmet acceptance criteria, with live broker and worker checks still unperformed because server access was blocked. | 00:01:45.353907 |
| 30 Oct 2026 | 15:21:19.56563<br>15:26:23.092301 | Ran the Merrypak production database load, fixed the restore step so it locates the Postgres tools and reports failures honestly, then re-ran and verified the restored data and duplicate-user cleanup. | 00:05:03.526671 |
| 02 Nov 2026 | 10:25:38.721088<br>10:25:39.53843 | Investigated the customer registration CAPTCHA report and updated the register page so a blocked or failed security check shows a loading state, a visible error with a reload option, and a clear submit message instead of an unsolvable prompt. | 00:00:00.817342 |
| 02 Nov 2026 | 10:25:39.53843<br>10:25:39.945 | Investigated the missing CAPTCHA on the Merrypak customer registration page and identified that the cookie consent banner covers the reCAPTCHA widget, plus a second failure when Google scripts are blocked. | 00:00:00.40657 |
| 02 Nov 2026 | 10:25:39.945<br>10:29:39.527 | Investigated the Merrypak registration page's reCAPTCHA, confirming the captcha renders but is initially unclickable because the cookie consent banner covers it, and verified the form blocks submission when Google's scripts are unavailable. | 00:03:59.582 |
| 02 Nov 2026 | 10:29:39.527<br>12:45:18.069807 | Investigated the missing CAPTCHA on the Merrypak customer registration page and identified that the cookie consent banner covers the reCAPTCHA widget, plus a second failure when Google scripts are blocked. | 02:15:38.542807 |
| 02 Nov 2026 | 12:45:18.069807<br>12:45:18.956431 | Investigated the customer registration CAPTCHA report and updated the register page so a blocked or failed security check shows a loading state, a visible error with a reload option, and a clear submit message instead of an unsolvable prompt. | 00:00:00.886624 |

## Merrypak

| Date | Start / end (SAST) | Work performed | Duration |
|---|---|---|---|
| 02 Nov 2026 | 12:45:18.956431<br>12:51:24.573252 | Investigated the missing CAPTCHA on customer registration and updated the register page so a blocked or slow security check shows a loading state, a visible failure message with a reload option, and a clearer submit error. | 00:06:05.616821 |
| 02 Nov 2026 | 12:51:24.573252<br>12:51:37.199585 | Investigated the customer registration CAPTCHA report and updated the register page so a blocked or failed security check shows a loading state, a visible error with a reload option, and a clear submit message instead of an unsolvable prompt. | 00:00:12.626333 |
| 02 Nov 2026 | 12:51:37.199585<br>12:52:19.807641 | Independently reviewed the registration CAPTCHA change and found it addresses a different failure cause than the reported one, leaves the invalid-key case undetected, and risks breaking slow mobile connections. | 00:00:42.608056 |
| 02 Nov 2026 | 12:52:19.807641<br>12:55:30.554727 | Investigated the customer registration CAPTCHA report and updated the register page so a blocked or failed security check shows a loading state, a visible error with a reload option, and a clear submit message instead of an unsolvable prompt. | 00:03:10.747086 |
| 02 Nov 2026 | 13:59:29.6632<br>13:59:30.558655 | Checked whether the registration CAPTCHA fix had been saved and shared, found it was still only a local uncommitted change, then saved and shared it after confirming the site builds and the copy check passes. | 00:00:00.895455 |
| 02 Nov 2026 | 13:59:30.558655<br>14:00:21.358662 | Reviewed the pending registration security-check fallback, checked its build and wording, and prepared the verified source change for release. | 00:00:50.800007 |
| 02 Nov 2026 | 14:00:21.358662<br>14:00:40.801575 | Checked whether the registration CAPTCHA fix had been saved and shared, found it was still only a local uncommitted change, then saved and shared it after confirming the site builds and the copy check passes. | 00:00:19.442913 |
| 02 Nov 2026 | 14:00:40.801575<br>14:01:45.887411 | Reviewed the registration security-check fallback and identified slow-loading recovery, timeout and accessibility issues that need correction. | 00:01:05.085836 |
| 02 Nov 2026 | 14:01:45.887411<br>14:04:55.412003 | Checked whether the registration CAPTCHA fix had been saved and shared, found it was still only a local uncommitted change, then saved and shared it after confirming the site builds and the copy check passes. | 00:03:09.524592 |
| **TOTAL** | | | **03:38:26.798153** |

## Multisnack | R0, struck through on Crystal Logic

| Date | Start / end (SAST) | Work performed | Duration |
|---|---|---|---|
| 25 Oct 2026 | 12:48:42.96<br>12:49:35.993 | ~~Added a validation tool for schedule-upload files and documented the division mapping.~~ | 00:00:53.033 |
| 30 Oct 2026 | 11:23:16.127<br>12:00:49.776 | ~~Reviewed a Multisnack support call recording, created a Multisnack working folder with call notes, a plan and prep notes for three people, and investigated why jobs stay red in the DPP to draft a WhatsApp reply to Roland.~~ | 00:37:33.649 |
| 31 Oct 2026 | 15:46:37.959907<br>15:55:57.651768 | ~~Investigated the Multisnack Daily Production Plans app after the client reported no user account and a broken D3 link; found the D3 job upload failing on a bad time value, changed the upload to skip only the faulty row and report warnings, and confirmed the fix on production.~~ | 00:09:19.691861 |
| **TOTAL** | | | **00:47:46.373861** |

## Player1Sport

| Date | Start / end (SAST) | Work performed | Duration |
|---|---|---|---|
| 24 Oct 2026<br>25 Oct 2026 | 23:18:38.918<br>00:36:43.363 | Updated survey readiness scoring, preserved reports when surveys are deleted, and added response dates; adjusted sprint-distance chart labels. | 01:18:04.445 |
| 25 Oct 2026 | 04:15:42.295<br>04:28:13.084 | Drafted and revised a client email summarizing the Player1Sport survey repairs, added the greeting, and linked it to the original conversation for review. | 00:12:30.789 |
| 26 Oct 2026 | 11:08:03.472<br>11:16:45.089 | Reviewed the telemetry-upload ticket and confirmed hardware verification, rather than an outstanding software change, was blocking completion. | 00:08:41.617 |
| 26 Oct 2026 | 11:38:49.309<br>11:41:13.555 | Checked the outstanding session-report and readiness tickets against implementation records and identified completed or externally blocked items. | 00:02:24.246 |
| 26 Oct 2026 | 11:51:44.841<br>11:54:53.777 | Reconciled the remaining ticket statuses and prepared a Jira update; direct posting was blocked because Jira was not connected. | 00:03:08.936 |
| 04 Nov 2026 | 14:40:12.446<br>14:54:52.796 | Created a 27-page current-system specification for Player1Sport, detailing application components, integrations, data flows, and database design based on existing code. | 00:14:40.35 |
| **TOTAL** | | | **01:59:30.383** |

## Snuza

| Date | Start / end (SAST) | Work performed | Duration |
|---|---|---|---|
| 28 Oct 2026 | 12:26:55.160898<br>12:28:41.42494 | Added a hidden honeypot field to the Snuza contact form to silently drop spam submissions and removed the hint text under the Send button, leaving the existing spam filter and CAPTCHA in place. | 00:01:46.264042 |
| 28 Oct 2026 | 14:36:06.989823<br>14:36:59.502271 | Verified the contact form honeypot spam filter and removal of the hint text under the Send button, confirmed the quoted signature block is not in the site code, and ran the contact page tests. | 00:00:52.512448 |
| 28 Oct 2026 | 14:41:25.574531<br>14:43:25.184096 | Re-verified the contact form honeypot spam filter and hint text removal, searched code and database dumps for the quoted signature block, and ran the contact page tests. | 00:01:59.609565 |
| 28 Oct 2026 | 15:14:29.364<br>15:19:23.034 | Merged the contact form spam and hint text changes into master, deleted the feature branch, and deployed the honeypot change to production, confirming the live contact page and spam drop behaviour. | 00:04:53.67 |
| 28 Oct 2026 | 16:56:07.148<br>17:00:55.044 | Investigated why the Snuza production checkout reported being behind, finding the code matched the main branch but history was broken, and drafted a reconciliation plan covering a missing documentation pack, test file gap and a broken static link. | 00:04:47.896 |
| 28 Oct 2026 | 17:41:49.784<br>17:55:16.522 | Reconciled the Snuza production checkout with the main code branch after backing it up, removed stale files and stashes, and fixed a broken static path without interrupting the live site. | 00:13:26.738 |
| 28 Oct 2026 | 18:15:53.372<br>19:13:13.474 | Diagnosed site freezes caused by blocking database calls stalling web workers, switched the worker type, replaced the restart cron with a health check, enabled HTTP/2 and compression, and raised API rate limits. | 00:57:20.102 |
| 28 Oct 2026 | 19:46:41.393<br>20:24:23.245 | Recovered and applied a missing database migration for the Snuza device API, deployed the pending API code, and ran the full endpoint test suite against production. | 00:37:41.852 |
| 28 Oct 2026 | 20:45:17.4<br>21:50:33.306 | Updated the website and device API communication helpers to isolate concurrent registration replies and handle connection failures without crashing. | 01:05:15.906 |

## Snuza

| Date | Start / end (SAST) | Work performed | Duration |
|---|---|---|---|
| 03 Nov 2026 | 16:44:23.614<br>18:06:31.1 | Removed the daily 04:15 Nginx reload and enabled global error logging, then reloaded Nginx and confirmed the site, API, login and registration checks still passed with no unexpected errors. | 01:22:07.486 |
| 06 Nov 2026 | 14:58:44.042<br>14:59:01.62 | Reviewed the Snuza website repository state and summarised the project setup, current branch status, local run steps and outstanding database restore flags. | 00:00:17.578 |
| 09 Nov 2026 | 09:02:21.086945<br>09:04:41.556159 | Investigated why the Snuza daily database backups stopped, finding no scheduled backup job on the server; the refresh ran but loaded the wrong dataset and skipped the production database. | 00:02:20.469214 |
| **TOTAL** | | | **04:32:50.083269** |

## Spoortrack [new]

| Date | Start / end (SAST) | Work performed | Duration |
|---|---|---|---|
| 05 Nov 2026 | 14:10:47.572<br>16:07:40.943 | Audited the logged-in app pages against the UI/UX system and applied a polish pass: centred the sidebar logo, removed repeated page titles, made the selected nav item a solid brand fill, unified toolbars, and added proper empty states and a clearer devices table. | 01:56:53.371 |
| 05 Nov 2026 | 16:56:06.113<br>17:15:22.244 | Polished the Spoortrack app interface: unified empty-state panels and widths, made all form buttons the app green, and added the public site navigation links into the in-app top bar. | 00:19:16.131 |
| 07 Nov 2026 | 10:08:31.693592<br>10:08:32.304005 | Ran the Spoortrack database loader, found it was importing a stale data set, published the pending archive and adjusted the publish schedule so daily sets are no longer a day late. | 00:00:00.610413 |
| 07 Nov 2026 | 10:08:32.304005<br>10:33:36.087827 | Ran the Spoortrack database load, found the newest published backup was a day stale, traced it to a publishing schedule that fires before the production copy finishes, published the pending set and adjusted the schedule. | 00:25:03.783822 |
| 07 Nov 2026 | 10:33:36.087827<br>10:33:38.2435 | Ran the Spoortrack database loader, found it was importing a stale data set, published the pending archive and adjusted the publish schedule so daily sets are no longer a day late. | 00:00:02.155673 |
| **TOTAL** | | | **02:41:16.051908** |

# Preserved calendar differences

These entries already existed in Tensology. They have not been edited or removed. "Outside reviewed" means the duration is not covered by the IDE-reviewed timeline in this report. It is a review difference, not an additional approved charge.

The three existing Spoortrack device-scheduling entries need a site and billability check against the current non-billable old-site classification. They remain exactly as requested.

Tensology entry 459 | Spoortrack [new] | 23 Oct 2026

Reorganised the device schedule page into separate grouped panels, fixed schedule interval heights, card rounding and spacing, aligned the GMT, GPS accuracy and VHF controls, and rebuilt the schedule styles from Sass with admin-page fixes.
15:20:00 - 16:16:00 SAST. Recorded: 00:56:00. IDE overlap: 00:00:00. Outside reviewed: 00:56:00.

Tensology entry 467 | Spoortrack [new] | 23 Oct 2026

Added spacing above the GPS readings block and tightened the GMT row on the device schedule page.
16:25:00 - 16:32:00 SAST. Recorded: 00:07:00. IDE overlap: 00:00:00. Outside reviewed: 00:07:00.

Tensology entry 468 | Spoortrack [new] | 23 Oct 2026

Made the GPS accuracy, on/off and VHF panels more compact, restored the Data Attributes allowance display, gave all panels the same background and vertically centred message boxes across the site.
16:46:00 - 17:09:00 SAST. Recorded: 00:23:00. IDE overlap: 00:00:00. Outside reviewed: 00:23:00.

Tensology entry 473 | Snuza | 28 Oct 2026

PROJ-169-001: Added contact-form spam honeypot protection, removed the obsolete Pico/signature hint, and verified the focused Snuza tests.
12:26:50 - 12:31:50 SAST. Recorded: 00:05:00. IDE overlap: 00:01:46.264042. Outside reviewed: 00:03:13.735958.

Tensology entry 461 | Merrypak | 23 Oct 2026

Moved D3 and Courier Guy credentials out of the code into environment settings, redacted customer personal data in logs, and fixed value card registration bugs.
20:32:00 - 21:02:00 SAST. Recorded: 00:30:00. IDE overlap: 00:00:00. Outside reviewed: 00:30:00.

Tensology entry 472 | Merrypak | 28 Oct 2026

PROJ-175-151: Investigated recurring Celery worker-dead alert; isolated a possible broker-vhost mismatch from repository evidence. Live server checks were blocked and no production change was made.
10:52:29 - 11:37:29 SAST. Recorded: 00:45:00. IDE overlap: 00:04:57.693118. Outside reviewed: 00:40:02.306882.

Tensology entry 462 | Jackcat | 23 Oct 2026

Tightened PayFast payment security: each payment notification is now checked for a valid signature, the correct merchant and amount, a PayFast source address and confirmation from PayFast itself before it is accepted. Fixed the logging of failed payments, added the credit receipt, admin and fraud alert email templates, and made the catalogue pages keep working if the cache server (Redis) is unavailable.
21:02:00 - 21:37:00 SAST. Recorded: 00:35:00. IDE overlap: 00:00:00. Outside reviewed: 00:35:00.

Tensology entry 471 | Jackcat | 23 Oct 2026

Server security maintenance on the Jackcat server: upgraded Node.js from 18.19.1 to 22.23.4 to close a published security vulnerability, applied all outstanding Ubuntu security updates, restarted onto the patched kernel (6.8.0-147), and checked that all services came back up correctly.
21:27:00 - 22:18:00 SAST. Recorded: 00:51:00. IDE overlap: 00:00:00. Outside reviewed: 00:51:00.